

Voltammetry Data for Aquatic Pollutants Detection: A Comparative Study of Dimensionality Reduction Techniques

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Abstract—Water pollution is an urgent environmental issue because it can have serious impacts on human health and the balance of ecosystems. The increase in industrial and agricultural activities has led to a rise in hazardous pollutants in water sources. One type of pollutant that has recently become a concern is heavy metals. Voltammetry methods have proven effective in detecting pollution, especially heavy metals in water. The data generated by voltammetry are complex and high-dimensional, necessitating procedures to reduce the dimensions. This research aims to find the best combination of dimensionality reduction techniques and classification methods. The dimensionality reduction techniques used are Principal Component Analysis (PCA), Linear Discriminant Analysis (LDA), AutoEncoder, and Independent Component Analysis (ICA). The classification methods implemented are Random Forest (RF), Support Vector Machine (SVM), K-Nearest Neighbors (KNN), Logistic Regression (LR), and Decision Tree (DT). The results of the study show that RF-PCA is the best combination, achieving an accuracy of 0.96.

Keywords—classification, dimensionality reduction, PCA, voltammetry, water pollution.

I. INTRODUCTION

Water pollution is an urgent environmental issue because it can have serious impacts on human health and ecosystem balance [1]. The increase in industrial and agricultural activities has led to a rise in hazardous pollutants in water sources [2]. One type of pollutant that has recently become a concern is heavy metals. Heavy metal contaminants including lead (Pb), cadmium (Cd), arsenic (As), mercury (Hg), and others can have very dangerous effects if present in the water used daily. Long-term exposure to heavy metals can cause various health problems, including organ damage, nervous system disorders, cancer, and developmental issues in children [3, 4]. Therefore, early and accurate detection of heavy metal pollution in water is essential.

One effective method for detecting pollutants in water is voltammetry. Voltammetry is an electrochemical analysis technique used to measure the content of chemical substances in a sample by observing the current changes produced by redox reactions at the electrode [5]. Voltammetry has high sensitivity and can detect various types of pollutants in water [6]. However, the voltammetry data generated is often very complex and high-dimensional [7, 8], making further analysis difficult.

To address this complexity, various dimensionality reduction techniques can be applied. Techniques for reducing dimensionality help streamline the data while preserving the key information [9, 10] necessary for pollutant detection. Commonly used techniques include Principal Component Analysis (PCA), Linear Discriminant Analysis (LDA), AutoEncoder, and Independent Component Analysis (ICA) [11–13].

In this study, various dimensionality reduction techniques will be compared to find the most effective method for simplifying voltammetry data. These dimensionality reduction techniques will then be combined with several classification algorithms, such as Random Forest (RF), Support Vector Machine (SVM), K-Nearest Neighbors (KNN), Logistic Regression (LR), and Decision Tree (DT). The combination of dimensionality reduction techniques and classification algorithms is expected to provide accurate results in detecting water pollutants.

Several previous studies have used voltammetry to detect pollutants in water. One such study was conducted by In a study by Jersson X. Leon-Medina *et al.* [14], support vector machines (SVM) were used for classifying heavy metal ions like As, Pb, and Cd. Dimensionality reduction was then performed using t-distributed stochastic neighbor embedding (t-SNE). The results of this study showed a high classification accuracy of 98.31%. Although the results are promising, this study only used SVM for classification and t-SNE for

dimensionality reduction, without exploring other potentially more effective combinations of techniques.

Another study by Jia-Jia Ye *et al.* [15] applied support vector machines (SVM) to classify electroanalytical signals of six heavy metal ions, using principal component analysis (PCA) to reduce the data dimensions based on their voltammetric peak shapes. This study used PCA for dimensionality reduction and SVM, KNN, Tree, and ensemble approaches as classifiers. The results of this study showed an accuracy rate of 90%. Although this study tried several classification methods, it did not explore other dimensionality reduction methods besides PCA.

Although these studies have shown promising results in detecting pollutants, especially heavy metals, using voltammetry and machine learning, there are still some limitations. These limitations are related to the variation in dimensionality reduction techniques and classification algorithms used. This study aims to fill this gap by evaluating the best combination of dimensionality reduction techniques and classification algorithms for detecting water pollutants from voltammetry data. The evaluation will be conducted using accuracy, precision, recall, and f1-score metrics [16]. The performance of each technique and algorithm combination will be measured to identify the most effective method combination for analyzing voltammetry data.

This paper is structured into several sections. Chapter 1 explains the background of the problem and the research objectives. Chapter 2 describes the research methodology. Chapter 3 presents the experimental results and performance comparison analysis. Chapter 4 presents the conclusions of the research findings and recommendations for further research.

II. METHODOLOGY

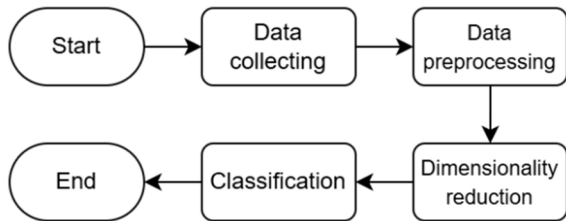


Fig. 1. Research stages

This research is conducted through several stages, which are illustrated in Fig. 1. Broadly, the research stages include the collection of voltammetry data, data preprocessing, application of dimensionality reduction techniques, and classification using various algorithms. Each of these stages will be explained in more detail in the following sub-chapters.

A. Data Description and Preprocessing

The data used in this research consists of voltammetry data in Excel format, obtained from the output of a potentiostat. The dataset includes 90 samples divided into three classes: two classes of heavy metals (Pb/Lead and Cd/Cadmium) and one class of non-heavy metals. Each class contains 30 samples.

Each voltammetry sample comprises two main features: voltage (V) and current (μA), each having approximately 800 rows. These data are then flattened, resulting in each sample having a dimensionality of 1600 features. However, not all

samples have 800 rows; some have fewer rows. To ensure uniformity, the data is cropped to 1500 features.

Once the samples are flattened and cropped, they are combined to form a data frame with a size of 90×1500 , where 90 represents the number of samples and 1500 represents the number of features. This preprocessing stage ensures that all samples have a consistent format for subsequent analysis.

B. Dimensionality Reduction Techniques

This research explores several dimensionality reduction techniques, namely: PCA, LDA, AutoEncoder, and ICA.

1) Principal Component Analysis (PCA)

Principal Component Analysis (PCA) is used to simplify data by converting it into a new set of variables, known as principal components [17]. These components are formed by linearly combining the original variables and are organized in a specific sequence. The advantage of PCA lies in its ability to reduce the data dimensions while retaining as much variance as possible, making it a popular initial step before applying other machine learning algorithms. PCA can be calculated using Eq. (1).

$$Z = WX \quad (1)$$

Where:

- Z is the transformed data matrix
- X is the original data matrix
- W is the matrix of eigenvectors from the covariance matrix of X

2) Linear Discriminant Analysis (LDA)

Linear Discriminant Analysis (LDA) is a technique used to find the linear combinations of features that best separate two or more classes of objects or events [18]. The goal of LDA is to maximize the ratio of between-class variance to within-class variance, ensuring optimal separation between the classes.

3) AutoEncoder

An AutoEncoder is a model designed to learn efficient, lower-dimensional representations of data [19, 20]. It consists of two main components: an encoder that transforms the input into a compact form, and a decoder that reconstructs the input from this condensed representation [21]. The advantage of AutoEncoders lies in their ability to learn non-linear representations of data, which provides high flexibility and power in reducing the dimensions of complex datasets.

4) Independent Component Analysis (ICA)

Independent Component Analysis (ICA) is a statistical technique used to identify independent components within data [22]. The advantage of ICA lies in the ability to uncover independent structures within data, which proves particularly useful in applications that require identifying hidden sources that are not dependent on each other. ICA can be calculated using Eq. (2).

$$X = AS \quad (2)$$

Where:

- X is the observed data matrix,
- A is the mixing matrix,
- S is the matrix of independent.

C. Classification Algorithm

In this study, several machine learning methods are employed, including RF, SVM, KNN, LR, and DT. These methods are evaluated to determine their effectiveness in combination with dimensionality reduction techniques for detecting water pollutants.

1) Random Forest (RF)

Random Forest (RF) is a method that enhances classification performance by aggregating predictions from multiple decision trees [23]. RF constructs several decision trees during training and assigns the class that most frequently appears in the predictions of the individual trees. RF is particularly advantageous due to its ability to prevent overfitting and its robustness against outliers. RF can be calculated using Eq. (3).

$$\hat{y} = \text{mode}(\hat{y}_1, \hat{y}_1, \dots, \hat{y}_n) \quad (3)$$

Where:

- $\hat{y}$ is the final predicted class
- $\hat{y}_1$ represents the prediction of the i -th tree.

2) Support Vector Machine (SVM)

Support Vector Machine (SVM) is a classification algorithm that finds the optimal hyperplane that maximizes the margin between two classes [24]. SVM works by transforming data into a higher dimension using the kernel trick. The advantage of SVM is its performance with unstructured and high-dimensional data. SVM can be calculated using Eq. (4).

$$f(x) = \text{sign}(w \cdot x + b) \quad (4)$$

Where:

- $f(x)$ represents the output classifier for a given input.
- w is the weight vector.
- x is the input vector.
- b is the bias term.

3) K-Nearest Neighbors (KNN)

K-Nearest Neighbors (KNN) is a method that assigns a label to a sample by considering the labels of its k closest neighbors and selecting the most frequent one [25]. The advantage of KNN is its simplicity and its ability to classify without making assumptions about the data distribution. KNN involves calculating the distance between points, typically using Euclidean distance, as shown in Eq. (5).

$$d(x, y) = \sum_{i=1}^n (x_i - y_i)^2 \quad (5)$$

Where:

- $d(x, y)$ represents the distance between two points
- x dan y are two data points
- x_i dan y_i are the values of the i -th dimension of x dan y
- n is the number of dimensions

4) Logistic Regression (LR)

Logistic Regression is a statistical technique used to model the relationship between one or more independent variables and a binary outcome. By applying the logistic function, it estimates the probability of a certain event occurring, providing a way to predict the likelihood of outcomes based on the input variables. This method is commonly used for classification tasks where the result is categorical, such as determining whether an event will happen or not [26]. The advantage of Logistic Regression lies in the model interpretability and efficiency in handling binary classification problems. Logistic Regression can be calculated using Eq. (6).

$$P(y = 1 | x) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_1 + \dots + \beta_n x_n)}} \quad (6)$$

Where:

- $P(y = 1 | x)$ is the probability of the event occurring.
- β_0 is the intercept term
- $\beta_1, \dots, \beta_n$ are the coefficients for each feature $x_1, \dots, x_n$

5) Decision Tree (DT)

Decision Tree is a machine learning algorithm that uses a tree-like structure of decisions [27] and their possible consequences. A decision tree is constructed by splitting the dataset into subsets based on attributes until reaching leaves representing the class labels. The advantage of Decision Tree lies in the ability to handle non-linear data and model interactions between features.

III. RESULT AND DISCUSSION

A. Dimensionality Reduction Result

In this study, several dimensionality reduction techniques were applied to the dataset, with each method yielding different optimal dimensions.

To determine the optimal dimensions from Principal Component Analysis (PCA), this study used the elbow method. The elbow method is a technique for identifying the optimal number of components by observing the point where adding more components no longer significantly increases the explained variance. The elbow method calculates the cumulative explained variance for each reduced dimension. This study defined the desired cumulative explained variance as 0.95, meaning the dimensionality reduction should still explain 95% of the variance in the data. Fig. 2 shows the results of the elbow method for PCA. The results from the elbow method indicate that the optimal number of components for PCA is 8.

Linear Discriminant Analysis (LDA) determines the optimal dimensions based on classification accuracy using the K-Nearest Neighbors (KNN) algorithm. LDA can only reduce dimensions up to the number of classes minus one. Therefore, for a dataset with three classes, LDA can only reduce dimensions to 1 or 2. KNN was chosen due to its simple concept in classification. The classification results showed that LDA with a two-dimensional reduction provided the best accuracy, making the optimal number of dimensions for LDA is 2.

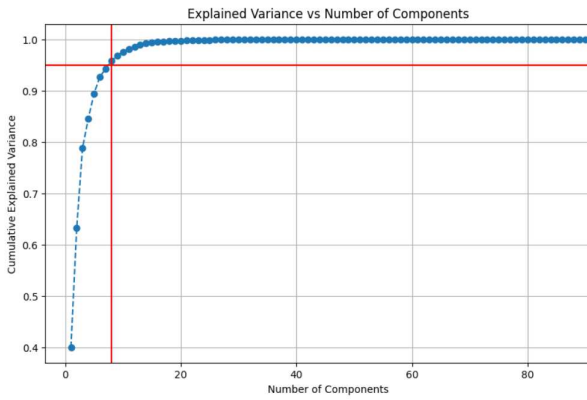


Fig. 2. Elbow method for determining the optimal dimension of PCA

For Independent Component Analysis (ICA), the optimal number of dimensions was determined by testing classification accuracy using KNN across a range of reduced dimensions from 1 to 101. The highest classification accuracy was achieved with 10 dimensions. Therefore, the optimal number of dimensions for ICA is 10.

Unlike LDA and ICA, the AutoEncoder determines the optimal dimensions based on reconstruction loss, where lower loss indicates better performance. Similar to ICA, the AutoEncoder was tested with dimensions ranging from 1 to 101, and the optimal dimension with the lowest reconstruction loss was found to be 81. A summary of the optimal dimensions obtained from each dimensionality reduction technique is presented in Table I.

B. Classification Result

After dimensionality reduction, the dataset was classified using the reduced dimensions obtained from PCA, LDA, ICA, and AutoEncoder. Each dimensionality reduction technique was combined with several classification methods. The classification methods used included RF, SVM, KNN, LR, and DT. The classification performance was evaluated to determine the effectiveness of the combination of dimensionality reduction techniques and classification methods.

The performance of the classification model was assessed using several metrics, including accuracy, precision, recall, and f1-score. To strengthen the reliability of the results, a stratified k-fold cross-validation method with $k=5$ was employed. In this method, the dataset was divided into 5 subsets, ensuring that each subset maintained the same class distribution as the original dataset. Each fold was used

TABLE I. OPTIMAL DIMENSIONS OF EACH DIMENSIONALITY REDUCTION TECHNIQUE

Dimensionality Reduction Techniques	Optimal Dimension
PCA	8
LDA	2
AutoEncoder	81
ICA	10

alternately as test data, while the other four folds were used as training data. The performance of each method combination can be seen in Table II.

In Table II, accuracy ("Acc") is shown as the percentage of correct predictions. Precision ("P") refers to the proportion of positive predictions that are truly correct. Recall ("R") measures how well the actual positive cases are identified, and the f1-score ("f1") is the harmonic average of precision and recall. The combination of RF and PCA yielded the best results, achieving 0.96 in all metrics. This combination produced the best performance likely because Random Forest can leverage the rich information from the eight principal components selected by PCA and has internal mechanisms to handle overfitting by using multiple decision trees. The classification report and confusion matrix of RF-PCA combination can be seen in Fig. 3 and Fig. 4.

	precision	recall	f1-score	support
Cd (Cadmium)	0.93	0.93	0.93	30
Non-Heavy Metal	0.97	0.93	0.95	30
Pb (Timbal)	0.97	1.00	0.98	30
accuracy			0.96	90
macro avg	0.96	0.96	0.96	90
weighted avg	0.96	0.96	0.96	90

Fig. 3. Classification report of RF-PCA

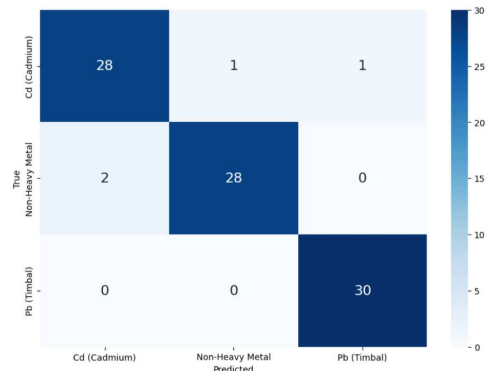


Fig. 4. Confusion matrix of RF-PCA

TABLE II. THE PERFORMANCE OF EACH METHOD COMBINATION

Classifier	PCA				LDA				AutoEncoder				ICA			
	Acc	P	R	f1	Acc	P	R	f1	Acc	P	R	f1	Acc	P	R	f1
RF	0.96	0.96	0.96	0.96	0.84	0.85	0.84	0.85	0.87	0.88	0.87	0.87	0.87	0.87	0.87	0.87
SVM	0.81	0.85	0.81	0.81	0.80	0.82	0.80	0.80	0.66	0.68	0.66	0.66	0.83	0.84	0.83	0.83
KNN	0.79	0.79	0.79	0.79	0.83	0.84	0.83	0.84	0.76	0.78	0.76	0.76	0.84	0.89	0.84	0.85
LR	0.86	0.86	0.86	0.86	0.78	0.80	0.78	0.78	0.91	0.92	0.91	0.91	0.82	0.83	0.82	0.82
DT	0.83	0.85	0.83	0.83	0.76	0.75	0.76	0.75	0.82	0.83	0.82	0.82	0.72	0.72	0.72	0.72

Conversely, the worst performance was achieved by the combination of SVM and AutoEncoder, with accuracy, recall, and f1-score at 0.66, and precision at 0.68. The likely cause of this poor performance is that SVM struggles to handle the 81 dimensions resulting from AutoEncoder, which might contain a lot of noise or irrelevant information, thus reducing the model's

IV. CONCLUSION

Voltammetry generates high-dimensional data, necessitating dimensionality reduction techniques to facilitate data processing and analysis. This research explores several dimensionality reduction methods combined with classification methods. Each data reduction method produces a different optimal number of dimensions. After data reduction, classification was performed with several methods, and the performance of each combination was evaluated using accuracy, precision, recall, and f1-score metrics. This study shows that the best performance was achieved by the combination of Random Forest (RF) and Principal Component Analysis (PCA), with accuracy reaching 0.96.

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REFERENCES

- [1] G. I. Edo, L. O. Itoje-akpokiniowo, P. Obasohan, V. O. Ikpekor, P. O. Samuel, A. N. Jikah, L. C. Nosu, H. A. Ekokotu, U. Ugbune, and E. E. A. Oghoro, "Impact of environmental pollution from human activities on water, air quality and climate change," *Ecological Frontiers*, 2024.
- [2] B. S. Rath, P. S. Kumar, and D.-V. N. Vo, "Critical review on hazardous pollutants in water environment: Occurrence, monitoring, fate, removal technologies and risk assessment," *Science of the Total Environment*, vol. 797, p. 149134, 2021.
- [3] M. H. Lafta, A. Afra, I. Patra, A. T. Jalil, M. J. Mohammadi, A. M. Baqir Al-Dhalimy, S. Ziyadullaev, F. Kiani, H. A. Ekrami, and P. Asban, "Toxic effects due to exposure heavy metals and increased health risk assessment (leukemia)," *Rev Environ Health*, vol. 39, no. 2, pp. 351–362, 2024.
- [4] C. R. Kothapalli, "Differential impact of heavy metals on neurotoxicity during development and in aging central nervous system," *Curr Opin Toxicol*, vol. 26, pp. 33–38, 2021.
- [5] S. Sawan, R. Maalouf, A. Errachid, and N. Jaffrezic-Renault, "Metal and metal oxide nanoparticles in the voltammetric detection of heavy metals: A review," *TrAC Trends in Analytical Chemistry*, vol. 131, p. 116014, 2020.
- [6] Y. Liu, Q. Xue, C. Chang, R. Wang, Z. Liu, and L. He, "Recent progress regarding electrochemical sensors for the detection of typical pollutants in water environments," *Analytical Sciences*, vol. 38, no. 1, pp. 55–70, 2022.
- [7] M. Kang, D. Kim, J. Kim, N. Kim, and S. Lee, "Strategies to Enrich Electrochemical Sensing Data with Analytical Relevance for Machine Learning Applications: A Focused Review," *Sensors*, vol. 24, no. 12, p. 3855, 2024.
- [8] K. Głowacz, U. E. Wawrzyniak, and P. Ciosek-Skibińska, "Comparison of various data analysis techniques applied for the classification of oligopeptides and amino acids by voltammetric electronic tongue," *Sens Actuators B Chem*, vol. 331, p. 129354, 2021.
- [9] S. Ayesha, M. K. Hanif, and R. Talib, "Overview and comparative study of dimensionality reduction techniques for high dimensional data," *Information Fusion*, vol. 59, pp. 44–58, 2020.
- [10] W. Jia, M. Sun, J. Lian, and S. Hou, "Feature dimensionality reduction: a review," *Complex & Intelligent Systems*, vol. 8, no. 3, pp. 2663–2693, 2022.
- [11] P. Li, Y. Pei, and J. Li, "A comprehensive survey on design and application of autoencoder in deep learning," *Appl Soft Comput*, vol. 138, p. 110176, 2023.
- [12] M. Ramamurthy, Y. H. Robinson, S. Vimal, and A. Suresh, "Auto encoder based dimensionality reduction and classification using convolutional neural networks for hyperspectral images," *Microprocess Microsyst*, vol. 79, p. 103280, 2020.
- [13] F. Anowar, S. Sadaoui, and B. Selim, "Conceptual and empirical comparison of dimensionality reduction algorithms (pca, kpca, lda, mds, svd, lle, isomap, le, ica, t-sne)," *Comput Sci Rev*, vol. 40, p. 100378, 2021.
- [14] J. X. Leon-Medina, D. A. Tibaduiza, J. C. Burgos, M. Cuenca, and D. Vasquez, "Classification of As, Pb and Cd heavy metal ions using square wave voltammetry, dimensionality reduction and machine learning," *IEEE Access*, vol. 10, pp. 7684–7694, 2022.
- [15] J.-J. Ye, C.-H. Lin, and X.-J. Huang, "Analyzing the anodic stripping square wave voltammetry of heavy metal ions via machine learning: Information beyond a single voltammetric peak," *Journal of Electroanalytical Chemistry*, vol. 872, p. 113934, 2020.
- [16] M. Hasanah, R. A. Putri, M. Aidie, R. Putra, and T. Ahmad, "Analysis of Weight-Based Voting Classifier for Intrusion Detection System," *International Journal of Intelligent Engineering and Systems*, vol. 17, no. 2, pp. 190–200, 2024.
- [17] R. Sarno, K. Triyana, S. I. Sabilla, D. R. Wijaya, D. Sunaryono, and C. Fatichah, "Detecting pork adulteration in beef for halal authentication using an optimized electronic nose system," *IEEE Access*, vol. 8, pp. 221700–221711, 2020.
- [18] M. Malikah, R. Sarno, and S. I. Sabilla, "Ensemble Learning for Optimizing Classification of Pork Adulteration in Beef Based on Electronic Nose Dataset," *International Journal of Intelligent Engineering & Systems*, vol. 14, no. 4, 2021.
- [19] E. Pintelas, I. E. Livieris, and P. E. Pintelas, "A convolutional autoencoder topology for classification in high-dimensional noisy image datasets," *Sensors*, vol. 21, no. 22, p. 7731, 2021.
- [20] S. Kim, S.-H. Chu, Y.-J. Park, and C.-Y. Lee, "A tied-weight autoencoder for the linear dimensionality reduction of sample data," *Sci Rep*, vol. 14, no. 1, p. 26801, 2024.
- [21] C. Zhang, Y. Geng, Z. Han, Y. Liu, H. Fu, and Q. Hu, "Autoencoder in autoencoder networks," *IEEE Trans Neural Netw Learn Syst*, vol. 35, no. 2, pp. 2263–2275, 2022.
- [22] J. L. McConn, C. R. Lamoureux, S. Poudel, B. O. Palsson, and A. V. Sastry, "Optimal dimensionality selection for independent component analysis of transcriptomic data," *BMC Bioinformatics*, vol. 22, pp. 1–13, 2021.
- [23] R. A. Putri, S. I. Sabilla, and R. Sarno, "Optimal Feature Selection Algorithm (FSA) for Electronic Nose Signal," In: *2023 1st International Conference on Advanced Engineering and Technologies (ICONNET)*, 2023, pp. 310–315.
- [24] U. I. Shabrina, R. Sarno, R. N. E. Anggraini, A. T. Haryono, and A. F. Septiyanto, "Sentiment Analysis of Presidential Candidate Debates from YouTube Videos," In: *2024 IEEE International Conference on Artificial Intelligence and Mechatronics Systems (AIMS)*, 2024, pp. 1–6.
- [25] I. A. Sabilla, R. A. Irawan, R. Sarno, A. Al Fauzi, D. R. Wijaya, and R. Gunawan, "Classification of Male and Female Sweat Odor in the Morning Using Electronic Nose," In: *2021 3rd East Indonesia Conference on Computer and Information Technology (EIConCIT)*, Apr. 2021, pp. 320–324. doi: 10.1109/EIConCIT50028.2021.9431909.
- [26] A. Cemiloglu, L. Zhu, A. B. Mohammednour, M. Azarafza, and Y. A. Nanehkaran, "Landslide susceptibility assessment for Maragheh County, Iran, using the logistic regression algorithm," *Land (Basel)*, vol. 12, no. 7, p. 1397, 2023.
- [27] M. Bansal, A. Goyal, and A. Choudhary, "A comparative analysis of K-nearest neighbor, genetic, support vector machine, decision tree, and long short term memory algorithms in machine learning," *Decision Analytics Journal*, vol. 3, p. 100071, 2022.